MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY

INTEROFFICE COMMUNICATION

TO: Daniel Dell, Permits Section, Water Bureau

FROM: Sylvia Heaton, Surface Water Assessment Section, Water Bureau

DATE: January 3, 2008

SUBJECT: Cadillac Wastewater Treatment Plant - Permit Review Recommendations
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257

We have reviewed the subject permit application dated March 25, 2007. The facility discharges 3.2 million gallons per day (4.95 cubic feet per second [cfs]) of treated sanitary wastewater through outfall 001 to the Clam River in Wexford County. The flows calculated for the Clam River at this location are 2.7 cfs, 14 cfs, and 3.8 cfs for the 95 percent exceedance, harmonic mean, and 90Q10, respectively.

Sources of information used for this review include the permit application, the Surface Water Assessment Section (SWAS) facility file, the current permit, discharge monitoring reports, compliance sampling inspection data from 2004 and 2006, and information from the SWAS DISTOX database.

We recommend that no changes be made in the facility's final effluent limitations or frequency of analyses from the current permit for total phosphorus, total residual chlorine, pH, and zinc.

In addition, the total mercury monitoring required by the current permit demonstrates there continues to be reasonable potential for the exceedance of water quality standards for this parameter. Therefore, **we recommend a monthly average water quality-based effluent limit (WQBEL) of 10 nanograms per liter (2.7×10^{-4} pounds per day) for total mercury**. Based on the mercury data submitted with the permit application, complying with the WQBEL should be feasible. Monitoring shall be conducted monthly and samples analyzed using United States Environmental Protection Agency (USEPA) Method 1631. The existing Pollution Minimization Program requirements in the permit should be modified to reflect recent changes in the language (Attachment 1).

The facility is currently required to perform quarterly whole effluent toxicity monitoring. However, because the effluent shows reasonable potential to exceed the preliminary effluent limit, **we recommend a monthly chronic whole effluent toxicity limit of 1.1 chronic toxic units. The facility shall monitor the effluent on a monthly basis as outlined in the attached language (Attachment 2).**

The current permit includes a quarterly monitoring requirement for cadmium. After reviewing the sources of information used to evaluate the discharge of cadmium, **we recommend that the monitoring requirement be dropped from the permit**. The permit also requires that lindane be analyzed on quarterly basis as well. However, after reviewing the sources of information used to evaluate the discharge, **we recommend that monitoring for lindane be increased from quarterly to monthly. The discharge information continues to**

demonstrate that lindane is being discharged at levels that come close to exceeding the monthly average WQBEL for lindane. Samples should be analyzed using a quantification level of 0.01 micrograms per liter (ug/l).

Furthermore, based on data submitted with the NPDES permit application, **we recommend monitoring for selenium and cyanide amenable**. **Monitoring for these parameters should be conducted quarterly with 24-hour composite samples for selenium and grab samples for cyanide amenable. Samples should be analyzed using a USEPA approved method for selenium and USEPA approved method 01A-1677 for cyanide amenable. Quantification levels of 2 ug/l and 5 ug/l for selenium and cyanide, respectively, should be used with the analyses above**. Based on a letter of application amendment from the facility (dated September 20, 2007), **we recommend monitoring of methyl *tert* butyl ether. Monitoring of this parameter should be conducted quarterly with a grab/composite sample and analyzed using USEPA approved methods and a quantification level of 1 ug/l.**

If you have any questions regarding these recommendations, please contact me at 517-373-1320.

sh:rm
Attachments
cc: Michael Stifler, Cadillac District Supervisor, Water Bureau
Brenda Sayles/Facility File, Water Bureau

ATTACHMENT 1

FOR **REISSUANCE** WHEN CURRENT PERMIT **CONTAINS A MERCURY LIMIT**

	Maximum Monthly				**Maximum Monthly**					
Total Mercury (for permit cycle)	(report)	---	---	lbs/day	(report)	---	---	ng/l	**???**	Grab
	12-Month Rolling Average				**12-Month Rolling Average**					
Total Mercury (for permit cycle)	INSERTBOX Calculation	---	2.7 x 10^{-4}	lbs/day	10 **OR** site specific LCA			ng/l	**???**	

a. Final Effluent Limitation for Total Mercury KEEP ONLY FOR FACILITIES WITH A MERCURY LIMIT

The final limit for total mercury is the Level Currently Achievable (LCA) based on a multiple discharger variance from the water quality-based effluent limit of 1.3 ng/l, pursuant to Rule 323.1103(9) of the Water Quality Standards. Compliance with the LCA shall be determined as a 12-month rolling average. The 12-month rolling average shall be determined by adding the present monthly average result to the preceding 11 monthly average results then dividing the sum by 12. KEEP NEXT SENTENCE **ONLY** FOR FACILITIES WITH **INSUFFICIENT** DATA FOR MERCURY LIMIT DETERMINATION, OTHEWISE DELETE: For facilities with insufficient data needed to calculate the 12-Month Rolling Average, enter '***E**' on your monthly Discharge Monitoring Report (DMR) form until 12 months (or the equivalent of 12 months for quarterly monitoring) of monthly monitoring data have been obtained, then begin reporting the calculated 12-Month Rolling Average as required. For facilities with quarterly monitoring requirements for total mercury, quarterly monitoring shall be equivalent to 3 months of monitoring in calculating the 12-month rolling average. Facilities that monitor more frequently than monthly for total mercury must determine the monthly average result, which is the sum of the results of all data obtained in a given month divided by the total number of samples taken, in order to calculate the 12-month rolling average. If the 12-month rolling average for any CHOOSE ONE [month/quarter] is less than the LCA, the permittee will be considered to be in compliance for total mercury for that CHOOSE ONE [month/quarter], provided the permittee is also in full compliance with the Pollutant Minimization Program for Total Mercury, set forth in Part I.A.INSERTBOX.

The permittee may choose to demonstrate that an alternate site-specific LCA is appropriate and request a permit modification. Such request and supporting documentation shall be submitted in writing to the Department. Supporting documentation shall include a minimum of 12 samples taken over a 12 month period in accordance with EPA Method 1631. Upon approval, this permit may be modified in accordance with applicable laws and rules to incorporate the alternate site-specific LCA as the effluent limitation for total mercury.

KEEP ONLY FOR FACILITIES WITH A FINAL MERCURY LIMIT AND MONITORING MORE FREQUENT THAN QUARTERLY: After a minimum of 12 monthly data points have been collected, the permittee may request a reduction in the monitoring frequency if the data indicate that the 12-month rolling average mercury concentration is less than 5 ng/l. This request shall contain an explanation as to why the reduced monitoring is appropriate and shall be submitted to the Department. Upon receipt of written approval and consistent with such approval, the permittee may reduce the monitoring frequency for total mercury indicated in Part I.A.INSERTBOX of this permit. The Department may revoke the approval for reduced monitoring at any time upon notification to the permittee

b. Total Mercury Testing Requirements
The analytical protocol for total mercury shall be in accordance with EPA Method 1631, Revision E, "Mercury in Water by Oxidation, Purge and Trap, and Cold Vapor Atomic Fluorescence Spectrometry". The quantification level for total mercury shall be 0.5 ng/l, unless a higher level is appropriate because of sample matrix interference. Justification for higher quantification levels shall be submitted to the Department within 30 days of such determination.

The use of clean technique sampling procedures is strongly recommended. Guidance for clean technique sampling is contained in: EPA Method 1669, *Sampling Ambient Water for Trace Metals at EPA Water Quality Criteria Levels (Sampling Guidance)*, EPA-821-R96-001, July 1996. Information and data documenting the permittee's sampling and analytical protocols and data acceptability shall be submitted to the Department upon request.

Mercury PMP – for facilities with an EXISTING LIMIT and PMP

Pollutant Minimization Program for Total Mercury Keep for facilities WITH an existing limit and PMP

The goal of the Pollutant Minimization Program is to maintain the effluent concentration of total mercury at or below 1.3 ng/l. The permittee shall continue to implement the Pollutant Minimization Program approved on INSERTBOX, and modifications thereto, to proceed toward the goal. CHOOSE ONE OF THE FOLLOWING: The Pollutant Minimization Program includes the following: **OR** The permittee shall submit to the Department a modified Pollutant Minimization Program on or before INSERTBOX[60 days from the effective date of the permit] to include the following minimum requirements:
REVIEW CURRENT PERMIT TO DETERMINE IF PMP REQUIRED THE MINIMUM FREQUENCIES INDICATED.

a. an annual review and semi-annual monitoring of potential sources of mercury entering the wastewater collection system;

b. a program for quarterly monitoring of influent INSERTBOX KEEP OR DELETE SLUDGE and periodic monitoring of sludge for mercury; and

c. implementation of reasonable cost-effective control measures when sources of mercury are discovered. Factors to be considered include significance of sources, economic considerations, and technical and treatability considerations.

On or before March 31 of each year, the permittee shall submit a status report for the previous calendar year to the Department that includes 1) the monitoring results for the previous year, 2) an updated list of potential mercury sources, and 3) a summary of all actions taken to reduce or eliminate identified sources of mercury.

Any information generated as a result of the Pollutant Minimization Program set forth in this permit may be used to support a request to modify the approved program or to demonstrate that the Pollutant Minimization Program requirement has been completed satisfactorily.

A request for modification of the approved program and supporting documentation shall be submitted in writing to the Department for review and approval. The Department may approve modifications to the approved program (approval of a program modification does not require a permit modification), including a reduction in the frequency of the requirements under items a. & b. if the data indicate that the 12-month rolling average mercury concentration is less than 5 ng/l.

This permit may be modified in accordance with applicable laws and rules to include additional mercury conditions and/or limitations as necessary.

DEFINITION – NET CONCENTRATION

Net Concentration means the difference between the intake concentration of a pollutant and the final effluent concentration of the same pollutant. (i.e. Final Effluent Concentration – Intake Concentration = Net Concentration) as allowed for in Title 40 of the Code of Federal Regulations, Part 132, Appendix F, Procedure 5, Subsection D.

ATTACHMENT 2

Delayed Whole Effluent (Chronic) Toxicity Limits with Schedule of Compliance and option for TRE

For SWAS use

This language establishes a compliance schedule, TRE requirements, and a final limit where reasonable potential has been established for the discharge to exceed the chronic WET standard in R 323.1057(6). Use this language when:

- *Reasonable potential exists for an exceedance of the chronic WET standard.*
- *Discharge is existing (vs. new).*
- *A compliance schedule is warranted.*
- *TRE has not been completed.*

Retain pH control statement if effluent total ammonia N is >3 mg/l; otherwise delete.

Whole Effluent Toxicity Requirements USE THIS CLAUSE FOR A DELAYED CHRONIC LIMIT WITH OPTION FOR TRE

	Maximum Limits for Quantity or Loading				Maximum Limits for Quality or Concentration					
	Frequency		Sample							
Parameter	Monthly	7-Day	Daily	Units	Monthly	7-Day	Daily	Units	of Analysis	Type
Acute Toxicity										
Through INSERTBOX DATE		---	---	---	---	---	(report)	TU_A	Quarterly	24-Hr Composite
Beginning INSERTBOX DATE			---	---	---	---	---	1.0	TU_A	Monthly 24-Hr Composite
Chronic Toxicity										
Through INSERTBOX DATE		---	---	---	(report)	---	---	TU_C	Quarterly	24-Hr Composite
Beginning INSERTBOX DATE			---	---	1.1	---	TU_C	Monthly	24-Hr Composite	

Test species shall include fathead minnow **and** *Ceriodaphnia dubia*. Testing and reporting procedures shall follow procedures contained in EPA-821-R-02-013, "Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms (Fourth Edition)." INSERTBOX KEEP OR DELETE NEXT SENTENCE When the effluent ammonia nitrogen (as N) concentration is greater than 3 mg/l, the pH of the toxicity test shall be maintained at a pH of 8 standard units. The acute toxic unit value (TU_A) and chronic toxic unit value (TU_C) for **each species tested** shall be reported on the Discharge Monitoring Report (DMR). If multiple chronic toxicity tests for the same species are performed during the month, the maximum TU_A value and monthly average TU_C value for the species shall be reported. For **each species <u>not tested</u>**, the permittee shall enter **"*W"** on the DMR. Completed toxicity test reports for each test conducted shall be retained by the permittee in accordance with the requirements of Part II.B.5. of this permit and shall be available for review by the department upon request. KEEP OR DELETE NEXT SENTENCE After INSERTBOX <u>SPECIFY TIME INTERVAL</u> of toxicity testing and upon approval of the Department, the monitoring frequency may be reduced if the test data indicate that the toxicity requirements of Rule 323.1219 of the Michigan Administrative Code are consistently being met. After one (1) year of toxicity testing and upon approval of the Department, the chronic toxicity tests may be performed using the more sensitive species identified in the chronic toxicity database. If a more sensitive species cannot be identified, the chronic toxicity tests shall be performed with both species. Toxicity test data acceptability is contingent upon the validation of the test method by the testing laboratory. Such validation shall be submitted to the Department upon request.

KEEP OR DELETE FOLLOWING PARAGRAPH
The permittee shall conduct a Toxicity Reduction Evaluation (TRE). The objective of the TRE shall be to reduce the toxicity of the final effluent from monitoring point INSERTBOX to $\leq$ INSERTBOX chronic toxic unit (TU_C) and $\leq$ 1.0 acute toxic unit (TU_A) by INSERTBOX **DATE**. The following documents are available as guidance to reduce toxicity to acceptable levels: Phase I, EPA/600/6-91/005F (chronic), EPA/600/6-91/003 (acute); Phase II, EPA/600/R-92/080 (acute and chronic); Phase III, EPA/600/R-92/081 (acute and chronic); and Publicly Owned Treatment Works (POTWs), EPA/833B-99/002. Annual progress reports shall be submitted to the Department within 30 days of the completion of the last test of each annual cycle.

1) When monitoring shows persistent exceedance of the INSERTBOX TU_C limit or the 1.0 TU_A limit for effluent toxicity, the Department will determine whether the permittee must implement the toxicity control program requirements specified in 2) below.

2) Upon written notification by the Department, the following conditions apply. Within 90 days of the notification, the permittee shall implement a Toxicity Reduction Evaluation (TRE). The objective of the TRE shall be to reduce the toxicity of the final effluent from monitoring point INSERTBOX to $\leq$ INSERTBOX TU_C and $\leq$ 1.0 TU_A. The following documents are available as guidance to reduce toxicity to acceptable levels. Phase I, EPA/600/6-91/005F (chronic), EPA/600/6-91/003 (acute); Phase II, EPA/600/R-92/080 (acute and chronic); Phase III, EPA/600/R-92/081 (acute and chronic); and POTWs, EPA/833B-99/002. Annual progress reports shall be submitted to the Department within 30 days of the completion of the last test of each annual cycle.